

WU Yandong

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EDUCATION

The Hong Kong Polytechnic University

Bachelor of Engineering in Electrical Engineering, GPA: 3.54/4.3

Sept. 2024 - Sept. 2028

Hong Kong SAR, China

- Year 3, Undergraduate

- Minor in Electronic System & Internet of Things

- Student Member of the HKIE (Hong Kong Institution of Engineers)

TECHNICAL & PROJECT EXPERIENCE

- Robocon Competition | Core Member, PolyU Robotics EE Team | Dec 2025 – Jun 2026
 - Developed an STM32-based lower-level controller for a fully autonomous robot, incorporating PID motor group control, Mecanum-wheel chassis kinematics, and UART communication with the upper-level companion computer.
- Wheel-Legged robot simulation | Personal Repository | Jun 2026 - 3 Jul 2026
 - Simulated a wheel-legged robot in Webots, implementing the LQR control and VMC algorithm for self-balancing control.
 - Github Link : github.com/W-dongdong/Wheel_Leg_simulation-Webots
- STM32 BSP Library | Personal Repository | Mar 2026 - Jun 2026
 - Developed a lightweight Board Support Package library in C and C++ for STM32, including but not limited to CAN, I2C, and the Mahony algorithm for the IMU.
 - GitHub Link : github.com/W-dongdong/stm32_bsp_lib
- Champion, AI Car Racing (University-level Competition) | Aug 2026
 - Developed a YOLO-based, vision-guided maze-solving robot car on the Jetson Nano, completing the race in 18.88 s.
 - Github Link : github.com/W-dongdong/2026PolyU-AI-Robot-Car-Race/tree/master
- Second Runner-up, Course Project Competition | 2024
 - Designed the mechanical structure of a 3-phase generator using SolidWorks.

SKILLS

- Programming Languages: C, C++, Python
- Embedded Systems:
 - Bare-metal STM32 development
 - Familiar with the peripheral communication protocols such as CAN bus, I2C and UART
- Control system:
 - Motor control using PID and cascade PID algorithms
 - Have learned the Linear system, understand how to analyze the properties of a simple system
 - Self-learn the modern control theory and know the basic principle of LQR.

CURRENT EXPLORATION

- ROS : Learning ROS2 and get familiar with Linux environment
- Real-Time Operating Systems: Studying FreeRTOS basics for embedded multi-tasking

HONOURS

- Champion of the AI Car Racing Competition | Aug 2026
- Listed on the Dean's Honours List for outstanding academic performance | 2024/25, 2025/2026 academic year
- Second runner-up, Large Group Project Competition | 2024